Group 6 — Team Task Plan

Music Popularity Capstone · Genre: acoustic

We are working in pairs. Every part of the notebook has at least two people on it, so if one person is unavailable, the other keeps the work moving.

Our Members

- Martha Afful
- Emmanuel Dotse
- Rosemond Konadu Baafi
- Rahinatu Mohammed Lawal
- Ann Awuku Fafa
- Faith Ayorkor Mensah

Daily Schedule: Wednesday to Saturday

The parts build on each other, so we work through them in order, two parts a day.

Wednesday

- Part 1 — Load the data
- Part 2 — Clean the data
- Working today: **Martha Afful, Emmanuel Dotse, Ann Awuku Fafa**

Thursday

- Part 3 — Explore the data
- Part 4 — Predict popularity
- Working today: **Rosemond Konadu Baafi, Martha Afful, Rahinatu Mohammed Lawal, Faith Ayorkor Mensah**

Friday

- Part 5 — Spot the hits
- Part 6 — Find the moods
- Working today: **Ann Awuku Fafa, Rahinatu Mohammed Lawal, Martha Afful, Faith Ayorkor Mensah**

Saturday

- Part 7 — Tune and finish
- Part 8 — Write it up
- Working today: **Emmanuel Dotse, Rahinatu Mohammed Lawal, Rosemond Konadu Baafi, Martha Afful, Faith Ayorkor Mensah**

Group 6 Task Allocation by Notebook Section

Part 1 — Load the data

- 1.1 Imports — Martha Afful and Emmanuel Dotse
- 1.2 Select the genre for the team — Martha Afful and Emmanuel Dotse
- 1.3 Load the dataset into the notebook — Martha Afful and Emmanuel Dotse
- 1.4 Review the dataset structure and summary — Martha Afful and Emmanuel Dotse
- 1.5 Check whether the selected genre exists in the data — Martha Afful and Emmanuel Dotse
- 1.6 Filter the dataset to the chosen genre — Martha Afful and Emmanuel Dotse
- Write: three observations about the filtered dataset — Martha Afful and Emmanuel Dotse

Part 2 — Clean the data

- 2.1 Remove the unnecessary index column — Emmanuel Dotse and Ann Awuku Fafa
- 2.2 Check for duplicate tracks — Emmanuel Dotse and Ann Awuku Fafa
- 2.3 Handle missing values in the dataset — Emmanuel Dotse and Ann Awuku Fafa
- 2.4 Convert duration from milliseconds to minutes — Emmanuel Dotse and Ann Awuku Fafa
- 2.5 Investigate unusual values such as zero popularity and very short tracks — Emmanuel Dotse and Ann Awuku Fafa
- 2.6 Create the hit label for classification — Emmanuel Dotse and Ann Awuku Fafa
- Write: cleaning log and justification — Emmanuel Dotse and Ann Awuku Fafa

Part 3 — Explore the data

- Chart 1: Identify the artists with the most tracks in the genre — Rosemond Konadu Baafi, Martha Afful and Faith Ayorkor Mensah
- Interpret the artist dominance chart — Rosemond Konadu Baafi, Martha Afful and Faith Ayorkor Mensah
- Chart 2: Explore the popularity distribution in the genre — Rosemond Konadu Baafi, Martha Afful and Faith Ayorkor Mensah
- Interpret the popularity histogram — Rosemond Konadu Baafi, Martha Afful and Faith Ayorkor Mensah
- Chart 3: Analyse how musical features relate to each other — Rosemond Konadu Baafi, Martha Afful and Faith Ayorkor Mensah
- Interpret the correlation heatmap — Rosemond Konadu Baafi, Martha Afful and Faith Ayorkor Mensah
- Chart 4: Investigate one custom question about the genre — Rosemond Konadu Baafi, Martha Afful and Faith Ayorkor Mensah
- Interpret the custom chart — Rosemond Konadu Baafi, Martha Afful and Faith Ayorkor Mensah

Part 4 — Predict popularity

- 4.1 Choose the features and target variable for regression — Rahinatu Mohammed Lawal and Rosemond Konadu Baafi
- 4.2 Split the data into training and testing sets — Rahinatu Mohammed Lawal and Rosemond Konadu Baafi
- 4.3 Build the baseline linear regression model — Rahinatu Mohammed Lawal and Rosemond Konadu Baafi
- 4.4 Interpret the model coefficients — Rahinatu Mohammed Lawal and Rosemond Konadu Baafi
- 4.5 Build the random forest regression model — Rahinatu Mohammed Lawal and Rosemond Konadu Baafi
- 4.6 Compare model performance and plot actual vs predicted popularity — Rahinatu Mohammed Lawal and Rosemond Konadu Baafi
- Write: report on prediction results and real-world meaning — Rahinatu Mohammed Lawal and Rosemond Konadu Baafi

Part 5 — Spot the hits

- 5.1 Define the target variable without leaking popularity into the model — Ann Awuku Fafa and Rahinatu Mohammed Lawal

- 5.2 Split and scale the data for classification — Ann Awuku Fafa and Rahinatu Mohammed Lawal
- 5.3 Train logistic regression and decision tree models — Ann Awuku Fafa and Rahinatu Mohammed Lawal
- 5.4 Analyse the confusion matrix and compare errors — Ann Awuku Fafa and Rahinatu Mohammed Lawal
- Write: interpret the classification results and business implication — Ann Awuku Fafa and Rahinatu Mohammed Lawal

Part 6 — Find the moods

- 6.1 Select the mood features and scale them for clustering — Martha Afful, Ann Awuku Fafa and Faith Ayorkor Mensah
- 6.2 Choose the number of clusters using the elbow plot — Martha Afful, Ann Awuku Fafa and Faith Ayorkor Mensah
- 6.3 Fit the K-means model and visualize the mood groups — Martha Afful, Ann Awuku Fafa and Faith Ayorkor Mensah
- 6.4 Analyse the characteristics of each mood cluster — Martha Afful, Ann Awuku Fafa and Faith Ayorkor Mensah
- Write: name the mood groups and explain their meaning — Martha Afful, Ann Awuku Fafa and Faith Ayorkor Mensah

Part 7 — Tune and finish

- 7.1 Try different model settings to improve performance — Emmanuel Dotse and Rahinatu Mohammed Lawal
- 7.2 Compare the model before and after tuning — Emmanuel Dotse and Rahinatu Mohammed Lawal
- Write: explain whether tuning improved the model and why — Emmanuel Dotse and Rahinatu Mohammed Lawal

Part 8 — Write it up

- Prepare the final report with summary, data description, charts, models, and recommendations — Rosemond Konadu Baafi, Martha Afful and Faith Ayorkor Mensah
- Final review and submission checklist — Rosemond Konadu Baafi, and Faith Ayorkor Mensah

Final Pair Summary

- Martha Afful: Parts 1, 3, 6
- Emmanuel Dotse: Parts 1, 2, 7
- Rosemond Konadu Baafi: Parts 3, 4, 8
- Rahinatu Mohammed Lawal: Parts 4, 5, 7
- Ann Awuku Fafa: Parts 2, 5, 6
- Faith Ayorkor Mensah: Parts 3, 6, 8

How the Pairs Work

In each pair, one person leads and the other supports and acts as backup. This gives us:

- Continuity if one person cannot finish their part
- Regular communication between partners, who share responsibility for getting the task done
- A shared understanding of every section
- A smoother final report

Our Agreement

We work in pairs on every major part of the notebook. If one person cannot finish their share, their partner takes charge and makes sure it is done on time. Every member contributes properly and understands the work in their sections.